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The Editor-in-Chief
Quantum Machine Intelligence, Springer Nature

10 November 2026

Subject: Submission of the manuscript "Hybrid Quantum–Classical Pauli-Correlator Architecture for Image Reconstruction"

Dear Editor-in-Chief,

We submit the above manuscript for consideration in *Quantum Machine Intelligence*. It introduces the **Pauli-Correlator Autoencoder (PCAE)**, a hybrid quantum–classical model that combines matrix product state (MPS) compression, a six-qubit variational circuit, Hamiltonian regularisation and a classical neural decoder. The question we address is whether local Pauli expectation values and two-qubit correlations can serve as a structured latent representation from which classical images are rebuilt. The main contributions are:

1. Two six-qubit architectures, PCAE1D and PCAE2D, with MPS-based feature extraction and different circuit connectivity.
2. Per-image reconstruction on six datasets: CIFAR-10, MNIST, Fashion-MNIST and three medical imaging sets.
3. A group-averaged observable representation that is equivariant under the dihedral group D_4, verified numerically.
4. Trained models run on IBM Quantum processors without retraining the classical decoder, giving 25.90 to 31.52 dB PSNR in a five-image pilot.
5. Classical baselines, component ablations, noise studies and controlled tests of the pairwise observables, which map the capabilities and limits of the architecture.

We do not claim a quantum computational advantage. The work is a systematic study of how quantum observables can enter a classical learning pipeline, supported by quantitative comparisons and real-hardware experiments. Since it brings together tensor networks, quantum machine learning and hybrid models that run on present-day hardware, we consider it a good fit for the readership of the journal.

Thank you for considering our manuscript.

Sincerely,

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